Customer Churn & Revenue Analysis

Predictive & Prescriptive Analytics

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Project Overview

- Analyze customer behavior in e-commerce dataset
- Predict customer churn
- Forecast customer revenue
- Provide business recommendations

Dataset Overview

5000 customer records

Features include:

- Demographics (Age, Gender, City)
- Purchase data (Total Amount, Quantity)
- Behavior (Pages Viewed, Session Duration)
- Delivery & Ratings

Churn Prediction

Target variable: Churn (0 = Stay, 1 = Leave)

Models used:

- Logistic Regression
- Random Forest
- XGBoost

Churn Results

- Logistic Regression: 97.7%
- Random Forest: 100%
- XGBoost: 100%

- Tree-based models performed best

Churn Drivers

Top features:

- Monetary
- Frequency
- Avg Order Value
- Delivery Days
- Age

- Customer value and experience drive churn

Revenue Forecasting

Target: Total_Spend

Models:

- Linear Regression
- Random Forest Regressor

Revenue Results

Linear Regression:

- MAE: 1315
- R^2: 0.089

Random Forest:

- MAE: 1413
- R^2: 0.065

- Linear model performed better

Revenue Drivers

Top features:

- Avg Discount
- Age
- Delivery Days
- Pages Viewed
- Session Duration

Prescriptive Analytics

Customer Segmentation:

- Low Value
- Medium Value
- High Value

Business Insights

- Low-value customers → highest churn
- High-value customers → most loyal
- Medium-value → growth opportunity

Recommendations

- Retain high-value customers (VIP programs)
- Upsell medium-value customers
- Re-engage low-value customers
- Optimize discounts
- Improve delivery speed

Conclusion

- Machine learning effectively predicts churn
- Revenue forecasting supports planning
- Data-driven decisions improve business performance